

Probability and Statistics

Testing Hypothesis

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A **statistical hypothesis** is a conjecture about a population parameter. This conjecture may or may not be true.

- If we want to decide **whether a given coin is biased**, we formulate the hypothesis that the coin is fair, i.e., $p = 0.5$, where p is the probability of heads.
- If we want to decide **whether one procedure is better than another**, we formulate the hypothesis that there is no difference between the procedures

Such hypotheses are often called null hypotheses.

The **null hypothesis** states that there is **no difference** between a parameter and a specific value, or that there is no difference between two parameters. It is denoted by H_0 .

The **alternative hypothesis**, symbolized by H_1 , states **the existence of a difference** between a parameter and a specific value, or states that there is a difference between two parameters.

Any hypothesis that differs from a given null hypothesis is called an alternative hypothesis.

For example, if the null hypothesis is $p = 0.5$, possible alternative hypotheses are $p > 0.7$, or $p \neq 0.5$, or $p < 0.5$.

Procedures that enable us to decide whether to accept or reject hypotheses or to determine whether observed samples differ significantly from expected results are called **tests of hypotheses**

A researcher reports that the average salary of assistant professors is more than \$42,000. A sample of 30 assistant professors has a mean salary of \$43,260. Test the claim that assistant professors earn more than \$42,000 per year.

$H_0 : \mu = \$42000$ and $H_1 : \mu > \$42000$ (claim).

The mean lifetime of **100** bulbs produced by is **1570 hours** with a $\sigma = 120$ hours. Test the hypothesis $\mu = 1600$ against the alternative hypothesis $\mu \neq 1600$

$H_0 : \mu = 1600$ and $H_1 : \mu \neq 1600$

It is claimed that automobiles are driven on average more than 20,000 kilometers per year. To test this claim

$H_0 : \mu = 20,000$ & $H_1 : \mu > 20,000$

Type I error A type I error occurs if you reject the null hypothesis when it is true.

Type II error A type II error occurs if you do not reject the null hypothesis when it is false.

	H_0 true	H_0 false
Reject H_0	Error Type I	Correct decision
Do not reject H_0	Correct decision	Error Type II

The **maximum probability to risk a Type I error** is called the **level of significance** of the test.

The maximum probability of committing a type I error is denoted by α .
That is,

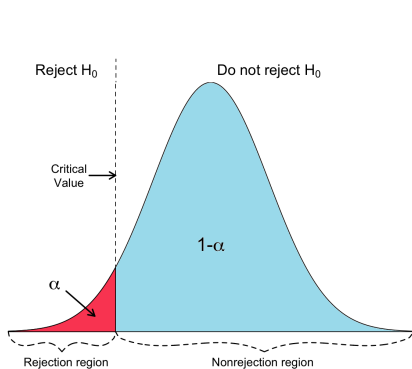
$$\alpha = P(\text{type I error}) = P\left(\frac{\text{reject } H_0}{H_0 \text{ is true}}\right)$$

In practice a level of significance of 0.05 or 0.01 is customary, although other values are used.

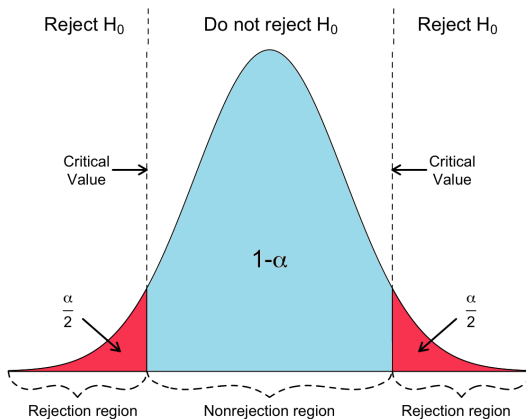
If for example a **0.05** or 5% level of significance is chosen in designing a test of a hypothesis, then there are about **5 chances in 100 that we would reject the hypothesis when it should be accepted**,

i.e., whenever the null hypotheses is true, we are about 95% confident that we would make the right decision.

In such cases we say that the hypothesis has been rejected at a 0.05 level of significance, which means that we could be wrong with probability 0.05.

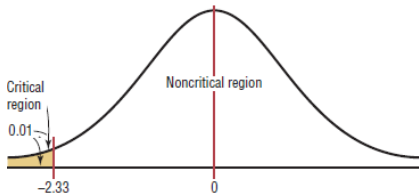


One tailed

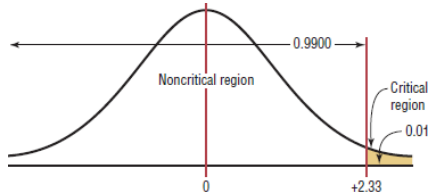


Two tailed

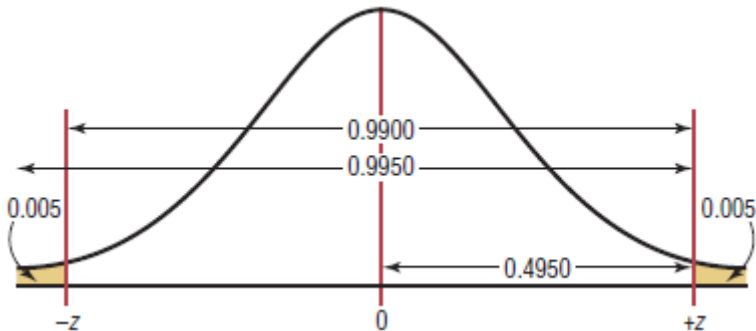
The **critical value** divides the area under the probability distribution curve in **rejection region(s)** and in **non-rejection region**.



$\alpha = 0.01$ (Left-Tailed Test)



$\alpha = 0.01$ (Right-Tailed Test)

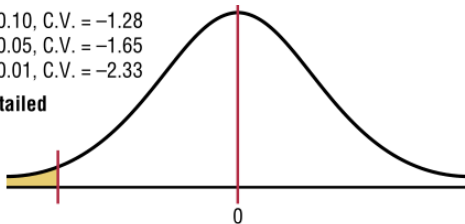


$\alpha = 0.01$ (Two-Tailed Test)

$$H_0: \mu = k \begin{cases} \alpha = 0.10, \text{C.V.} = -1.28 \\ \alpha = 0.05, \text{C.V.} = -1.65 \\ \alpha = 0.01, \text{C.V.} = -2.33 \end{cases}$$

$$H_1: \mu < k$$

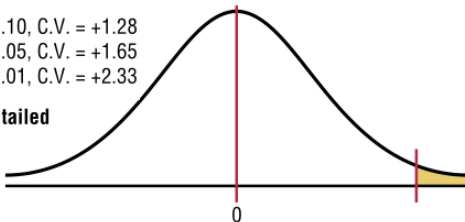
(a) Left-tailed



$$H_0: \mu = k \begin{cases} \alpha = 0.10, \text{C.V.} = +1.28 \\ \alpha = 0.05, \text{C.V.} = +1.65 \\ \alpha = 0.01, \text{C.V.} = +2.33 \end{cases}$$

$$H_1: \mu > k$$

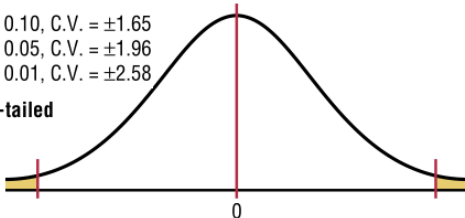
(b) Right-tailed



$$H_0: \mu = k \begin{cases} \alpha = 0.10, \text{C.V.} = \pm 1.65 \\ \alpha = 0.05, \text{C.V.} = \pm 1.96 \\ \alpha = 0.01, \text{C.V.} = \pm 2.58 \end{cases}$$

$$H_1: \mu \neq k$$

(c) Two-tailed



Significant Level and Critical Region

If the significance level α , then the critical region will consist of all values of Z which are

- (i) less than $-z_{\frac{\alpha}{2}}$ and greater than $z_{\frac{\alpha}{2}}$ in case of two-tailed test;

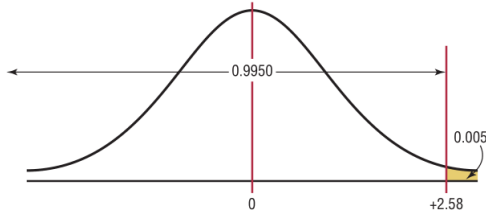
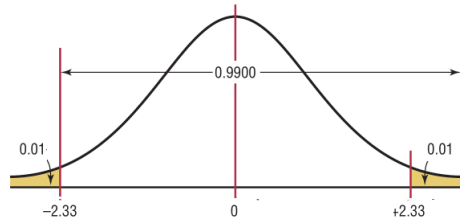
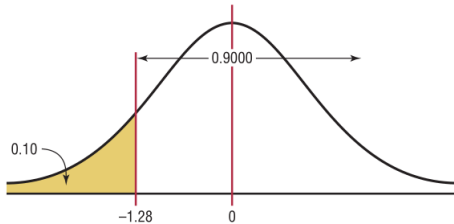
Significance level α	Two-tailed test	One-tailed test
0.10	± 1.645	± 1.28
0.05	± 1.96	± 1.645
0.01	± 2.58	± 2.33

Find the critical value(s) for each situation and draw the appropriate figure, showing the critical region.

- 1 A left-tailed test with $\alpha = 0.10$.
- 2 A two-tailed test with $\alpha = 0.02$.
- 3 A right-tailed test with $\alpha = 0.005$.

Find the critical value(s) for each situation and draw the appropriate figure, showing the critical region.

- ① A left-tailed test with $\alpha = 0.10$.
- ② A two-tailed test with $\alpha = 0.02$.
- ③ A right-tailed test with $\alpha = 0.005$.



Approach to Hypothesis Testing with Fixed Probability of Type I Error

1. State the null and alternative hypotheses.
2. Choose a fixed significance level α .
3. Choose an appropriate test statistic and establish the critical region based on α .
4. Reject H_0 if the computed test statistic is in the critical region. Otherwise, do not reject.
5. Draw scientific or engineering conclusions.

Significance Testing (P-Value Approach)

1. State the null and alternative hypotheses.
2. Choose an appropriate test statistic.
3. Compute the P-value based on the computed value of the test statistic.
4. Use judgment based on the P-value and knowledge of the scientific system.

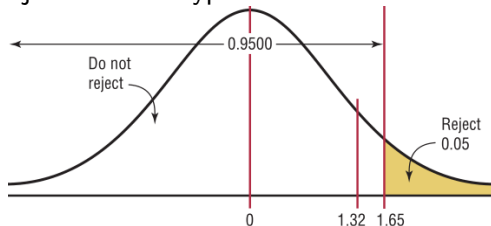
A researcher reports that the average salary of assistant professors is more than \$42,000. A sample of 30 assistant professors has a mean salary of \$43,260. At a $\alpha = 0.05$, test the claim that assistant professors earn more than \$42,000 per year. The standard deviation of the population is \$5230.

$H_0 : \mu = \$42000$ and $H_1 : \mu > \$42000$ (claim). At a $\alpha = 0.05$, $z = 1.65$.

z Test for mean

$$\text{Test value} = \frac{\text{Observed value} - \text{Expected value}}{\text{standard error}} \implies z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} = 1.32$$

Since $1.32 < 1.65$, and is not in the critical region, the decision is to not reject the null hypothesis.



Summarize the results. There is not enough evidence to support the claim that assistant professors earn more on average than \$42,000 per year.

The mean lifetime of **100** bulbs produced by is **1570 hours** with a $\sigma = 120$ hours.

Test the hypothesis $\mu = 1600$ against the alternative hypothesis $\mu \neq 1600$ ($\mu < 1600$), using a level of significance of 0.05

$H_0 : \mu = 1600$ and $H_1 : \mu \neq 1600$
(Two tailed Test). At a $\alpha = 0.05$,
 $z = \pm 1.96$.

(1) Reject H_0 if the z score of the sample mean is outside the range to 1.96.

(2) Accept H_0 otherwise.

$\mu = 1600, \sigma = 120$ and $n = 100$

gives $z = \frac{\bar{X} - 1600}{120/\sqrt{100}} = -2.50$ lies outside the range -1.96 to 1.96, we reject H_0 at a 0.05 level of significance.

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Test the hypothesis $\mu = 1600$ against the alternative hypothesis $\mu \neq 1600$ ($\mu < 1600$), using a level of significance of 0.05

$H_0 : \mu = 1600$ and $H_1 : \mu \neq 1600$

(Two tailed Test). At a $\alpha = 0.05$, $z = \pm 1.96$.

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$\mu = 1600, \sigma = 120$ and $n = 100$

gives $z = \frac{\bar{X} - 1600}{120/\sqrt{100}} = -2.50$ lies outside the range -1.96 to 1.96, we reject H_0 at a 0.05 level of significance.

$H_0 : \mu = 1600$ and $H_1 : \mu < 1600$
(One (left)tail Test). At a $\alpha =$

0.05, $z = -1.645$.

(1) Reject H_0 if the z is less than -1.645.

(2) Accept H_0 otherwise.

$\mu = 1600, \sigma = 120$ and $n = 100$

gives $z = \frac{\bar{X} - 1600}{120/\sqrt{100}} = -2.50$ less 1.645, we reject H_0 at a 0.05 level of significance.

The mean lifetime of **100** bulbs produced by is **1570 hours** with a $\sigma = 120$ hours.

Test the hypothesis $\mu = 1600$ against the alternative hypothesis $\mu \neq 1600$ ($\mu < 1600$), using a level of significance of **0.001**.

$H_0 : \mu = 1600$ and $H_1 : \mu \neq 1600$
(Two tailed Test). At a $\alpha = 0.001$, $z = \pm 2.58$.

(1) Reject H_0 if the z score of the sample mean is outside the range to 2.58.

(2) Accept H_0 otherwise.

$\mu = 1600$, $\sigma = 120$ and $n = 100$

gives $z = \frac{\bar{X} - 1600}{120/\sqrt{100}} = -2.50$ lies inside the range -2.58 to 2.58, we accept H_0 at **0.001** level of significance.

The mean lifetime of 100 bulbs produced by is 1570 hours with a $\sigma = 120$ hours.

Test the hypothesis $\mu = 1600$ against the alternative hypothesis $\mu \neq 1600$ ($\mu < 1600$), using a level of significance of 0.001.

$H_0 : \mu = 1600$ and $H_1 : \mu \neq 1600$

(Two tailed Test). At a $\alpha = 0.001$, $z = \pm 2.58$.

(1) Reject H_0 if the z score of the sample mean is outside the range to 2.58.

(2) Accept H_0 otherwise.

$\mu = 1600, \sigma = 120$ and $n = 100$

gives $z = \frac{\bar{X} - 1600}{120/\sqrt{100}} = -2.50$ lies inside the range -2.58 to 2.58, we accept H_0 at 0.001 level of significance.

$H_0 : \mu = 1600$ and $H_1 : \mu < 1600$
(One (left)tail Test). At a $\alpha = 0.001$, $z = -2.33$.

(1) Reject H_0 if the z is less than -2.33.

(1) Reject H_0 if the z is less than -2.33.

(2) Accept H_0 otherwise.

$\mu = 1600, \sigma = 120$ and $n = 100$

gives $z = \frac{\bar{X} - 1600}{120/\sqrt{100}} = -2.50$ less - 2.33, we reject H_0 at a 0.001 level of significance.

Decision Rule When Using a P-Value

- If $P\text{-value} \leq \alpha$, reject the null hypothesis.
- If $P\text{-value} > \alpha$, do not reject the null hypothesis.

There are two schools of thought on P-values. Some researchers do not choose an a value but report the P-value and allow the reader to decide whether the null hypothesis should be rejected.¹

- If $P\text{-value} \leq 0.01$, reject the null hypothesis. The difference is highly significant.
- If $P\text{-value} > 0.01$ but $P\text{-value} \leq 0.05$, reject the null hypothesis. The difference is significant.
- If $P\text{-value} > 0.05$ but $P\text{-value} \leq 0.10$, consider the consequences of type I error before rejecting the null hypothesis.
- If $P\text{-value} \geq 0.10$, do not reject the null hypothesis. The difference is not significant.

¹In this case, the following guidelines can be used, but be advised that these guide lines are not written in stone, and some statisticians may have other opinions.

The mean lifetime of **100** bulbs produced by is **1570 hours** with a $\sigma = 120$ hours. Test the hypothesis $\mu = 1600$ against the alternative hypothesis $\mu \neq 1600$, using a level of significance of **0.001** and find the P value of the test.

$$H_0 : \mu = 1600 \text{ and } H_1 : \mu \neq 1600,$$

$$\alpha = 0.001$$

$$\mu = 1600, \sigma = 120 \text{ and } n = 100 \text{ gives } z = \frac{\bar{X} \pm 1600}{120/\sqrt{100}} = \pm 2.50$$

$$P(-2.5 \leq z \leq 2.5) = P(z \leq 2.5) - P(z \geq -2.5) = \underbrace{0.0124}_{P\text{-value}}$$

$$P\text{-value} > \alpha$$

Since the **P-value is more than 0.001**, the decision is not to reject the null hypothesis.

Comparison Between α and P - value

$$H_0 : \mu = 1600 \text{ and } H_1 : \mu \neq 1600$$

$$\alpha = 0.001 \implies z_{\frac{\alpha}{2}} = \pm 2.58$$

$$\mu = 1600, \sigma = 120 \text{ and } n = 100$$

gives
$$z_o = \frac{\bar{X} - 1600}{120/\sqrt{100}} = -2.50$$

Reject H_0 if $z_o \notin [-z_{\frac{\alpha}{2}}, z_{\frac{\alpha}{2}}]$

Accept H_0 if $z_o \in [-z_{\frac{\alpha}{2}}, z_{\frac{\alpha}{2}}]$

Since $-2.5 \in [-2.58, 2.58]$, we accept H_0 at 0.001 level of significance.

$$H_0 : \mu = 1600 \text{ and } H_1 : \mu \neq 1600$$

$$\alpha = 0.001$$

$$\mu = 1600, \sigma = 120 \text{ and } n = 100$$

gives
$$z_o = \frac{\bar{X} - 1600}{120/\sqrt{100}} = -2.50$$

$$P(-2.5 \leq z \leq 2.5) = 0.0124$$

- If P-value $\leq \alpha$, reject H_0
- If P-value $> \alpha$, accept H_0 .

Since the P-value is more than 0.001, the decision is accept H_0 .

It is claimed that automobiles are driven on average more than 20,000 kilometers per year. To test this claim, 100 randomly selected automobile owners are asked to keep a record of the kilometers they travel. Would you agree with this claim if the random sample showed an average of 23,500 kilometers and a standard deviation of 3900 kilometers? Use a P-value in your conclusion.

$$H_0 : \mu = 20,000 \text{ \& } H_1 : \mu > 20,000$$

$$\bar{X} = 23,500; \sigma = 3900 \text{ and } n = 100$$

$$z_o = \frac{23,500 - 20,000}{\frac{3900}{\sqrt{100}}} = 8.97$$

$$P(\overset{\text{one tailed}}{\underbrace{z > z_0}}) = P(z > 8.97) \simeq 0$$

Rule

If $P\text{-value} \leq 0.01$, reject the null hypothesis. The difference is highly significant.

Yes I will agree with the claim that automobiles are driven on average more than 20,000 kilometers per year.

The average height of females in the freshman class of a certain college has historically been 162.5 centimeters with a standard deviation of 6.9 centimeters. Is there reason to believe that there has been a change in the average height if a random sample of 50 females in the present freshman class has an average height of 165.2 centimeters? Use a P-value in your conclusion. Assume the standard deviation remains the same.

To test the hypothesis that a coin is fair, using decision rules are adopted:

- (1) Accept the hypothesis if the number of heads in a single sample of 100 tosses is between 40 and 60 inclusive,
- (2) Reject the hypothesis otherwise.

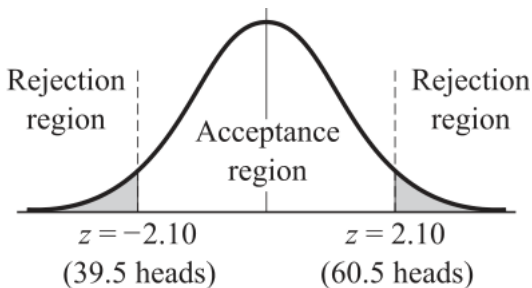
- (a) Find probability of rejecting the hypothesis when it is correct.
- (b) Interpret graphically the decision rule and the result of part (a).
- (c) Draw conclusions if 100 tosses yielded 53 heads? 60 heads?

The mean and standard deviation of the number of heads in 100 tosses are given by $\mu = np = 100 \left(\frac{1}{2}\right) = 50$ and $\sigma = \sqrt{npq} = 5$. As $np > 5$, $nq > 5$, the normal approximation to the binomial distribution can be used.

On a continuous scale, between 40 and 60 heads inclusive is the same as between 39.5 and 60.5 heads.

$$P(39.5 < X < 60.5) = P(-2.10 < Z < 2.10) = 0.9642$$

- (a) the probability of not getting between 40 and 60 heads inclusive if the coin is fair equals $1 - 0.9642 = 0.0358$. Then **the probability of rejecting the hypothesis when it is correct equals 0.0358**.
- (b)



- (c) According to the decision rule, we would have to accept the hypothesis that the coin is fair in **both** cases. If only one more head had been obtained, we would have rejected the hypothesis. This is what one must face when any sharp line of division is used in making decisions.

Design a decision rule to test the hypothesis that a coin is fair if a sample of 64 tosses of the coin is taken and if a level of significance of (a) 0.05, (b) 0.01 is used.

First Method:

If the level of significance is 0.05, it means that non-rejection region area is 95% implies that $-1.96 \leq z \leq 1.96$.

- (1) Accept the hypothesis that the coin is fair if $-1.96 \leq z \leq 1.96$.
- (2) Reject the hypothesis otherwise.

$$-1.96 \leq z \leq 1.96 \implies 24.16 \leq x \leq 39.84, \mu = np = 32, \sigma = \sqrt{npq} = 4$$

- (1) Accept the hypothesis that the coin is fair if the number of heads is between 24.16 and 39.84, i.e., between 25 and 39 inclusive.
- (2) Reject the hypothesis otherwise.

Second Method:

With probability 0.95, the number of heads will lie between $\mu - 1.96\sigma$ and $\mu + 1.96\sigma$. As

$$\mu = np = 64(0.5) = 32, \quad \text{and } \sigma = \sqrt{npq} = \sqrt{64(0.5)(0.5)} = 4.$$

Thus, with probability 0.95, the number of heads will lie between

$$32 - 1.96(4), \quad \text{and } 32 + 1.96(4) \implies 24.16 \text{ and } 39.84$$

- (1) Accept the hypothesis that the coin is fair if the number of heads is between 24.16 and 39.84, i.e., between 25 and 39 inclusive.
- (2) Reject the hypothesis otherwise.

A fabric manufacturer believes that the proportion of orders for raw material arriving late is $p = 0.6$. If a random sample of 10 orders shows that 3 or fewer arrived late, the hypothesis that $p = 0.6$ should be rejected in favor of the alternative $p < 0.6$. Use the binomial distribution. (a) Find the probability of committing a type I error if the true proportion is $p = 0.6$. (b) Find the probability of committing a type II error for the alternatives $p = 0.3$, $p = 0.4$, and $p = 0.5$.

$H_0 : p = 0.6$ and $H_1 : p < 0.6$. To test this hypothesis a random sample of $n = 10$ is selected if 3 or fewer arrived late, H_0 should be rejected.

$$\alpha = P(\text{Rejecting } H_0) = P(0 \leq X \leq 3) = \sum_{x=0}^3 C_x^{10} (0.6)^x (0.4)^{10-x} = 0.0548$$

(b)

$$H_0 : p = 0.6 \text{ and } H_1 : p < 0.3.$$

$$\beta = P \left(\begin{array}{c} \text{not rejecting } H_0 \\ \text{When it is false} \end{array} \right) = P \left(\begin{array}{c} 4 \text{ to } 10 \text{ orders} \\ \text{for raw material} \\ \text{arriving late} \end{array} \quad \text{when } p = 0.3 \right)$$

For $n = 10$ and $p = 0.3$, we have

$$\beta = \sum_{x=4}^{10} C_x^{10} (0.3)^x (0.7)^{10-x} = 1 - \sum_{x=0}^3 C_x^{10} (0.3)^x (0.7)^{10-x} = 0.3504$$

$$H_0 : p = 0.6 \text{ and } H_1 : p < 0.4.$$

For $n = 10$ and $p = 0.4$, we have

$$\beta = \sum_{x=4}^{10} C_x^{10} (0.4)^x (0.6)^{10-x} = 1 - \sum_{x=0}^3 C_x^{10} (0.4)^x (0.6)^{10-x} = 0.6177$$

$$H_0 : p = 0.6 \text{ and } H_1 : p < 0.5.$$

For $n = 10$ and $p = 0.5$, we have $\beta = 0.8281$.